

Jinwon Lee

SRE ENGINEER · PLATFORM ENGINEER

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"Make it work, then make it better."

Summary

I'm currently working as a Site Reliability Engineer at Samsung Electronics, with 8 years of experience as a developer. My background spans distributed AI systems, AI applications, infrastructure automation, and DevOps. I enjoy making life easier for developers by automating systems in Linux environments, and I'm always curious to dive deeper into how systems work. I'm also quick to explore and adopt new technologies when they can help solve real problems.

Work Experience

Samsung Research

Seoul, S.Korea

SRE(SITE RELIABILITY ENGINEER)

July. 2024 - Cur

- Architecting and operating a global-scale chatbot service on Microsoft Azure, collaborating with external partners and leading the system design and production readiness for multi-region deployment, initially launching in the UK and South Korea.
- Led the operation of a company-wide AI chatbot platform serving 30,000 employees, owning end-to-end infrastructure, platform, and application operations on Kubernetes while ensuring compliance with internal security, authentication, and data protection standards.
- Responsible for operating and provisioning development, staging, and production Kubernetes clusters, utilizing Infrastructure as Code (IaC) tools such as Helm and Helmfile.
- Built fully automated CI/CD pipelines for containerized applications using Docker, Kubernetes, Github Action and ArgoCD.
- Deployed a centralized monitoring environment(Grafana, Prometheus, Opentelemetry) which gather system metrics as well as docker run-time metrics.
- Provided personalized namespaces and essential modules in a Kubernetes cluster to offer user-friendly environments for development and experimentation, including dedicated test environments.
- Built an automated pipeline for evaluating the performance and quality of LLM models across Machine Translation.

Samsung Electronics (Network Dep.)

Seoul, S.Korea

SRE(SITE RELIABILITY ENGINEER) & SOFTWARE ENGINEER

Jan. 2021 - Jun. 2024

- Deployed containerized application using Docker, helm, and Operator(Implemented by OperatorSDK) and several AWS stack(ex. EKS, ECS, Route 53, RDS, IAM), focusing on high-availability, fault-tolerance, and auto-scaling.
- Experienced in building and managing diverse cloud platforms such as OpenShift, Kubernetes, and EKS, with a focus on optimizing cloud applications for performance and reliability in each environment.
- Developed an automated one-click provisioning system for Kubernetes clusters, providing cloud developers with streamlined, ready-to-use development environments.
- Developed a custom application to manage the extensive configuration parameters of a VM-based application, enabling users to validate and update settings through a familiar interface such as Excel.
- Built automated CI/CD systems leveraging Jenkins, GitHub Actions, and multiple cloud platforms to streamline development and deployment workflows.
- Developed an application that mirrors up to 130,000 packets per second in real time and forwards them to a target server, all within constrained system resources.

TmaxOS & TmaxCloud

Gyeonggi, S.Korea

SOFTWARE ENGINEER & PLATFORM ENGINEER

Jan. 2019 - Dec. 2020

- Developed CI/CD and registry operators in a Kubernetes environment to establish a fully automated pipeline, and deployed a private container image registry within the cloud.
- Built a Siri-like AI assistant on TmaxOS, focusing on system-level integration, AI model inference, and data preprocessing. Designed an asynchronous architecture to enable smooth and natural user experiences.

Education

UNIST(Ulsan National Institute of Science and Technology)

Ulsan, S.Korea

M.S. IN COMPUTER SCIENCE AND ENGINEERING

Mar. 2017 - Feb. 2019

- Advisor: Jiwon Seo (HanYang University)
- Advisor: Sam H. Noh (UNIST)
- Master Thesis : Convergence Aware CNN Training

- Minor: Mathematics
- Undergraduate Research Intern in Big Data Systems Lab (Prof : Jinwon Seo): Topic - Distributed system in AI model training(TensorFlow)

Publications

Poster	Alleviating the Network Bottleneck for CNN Distributed Training through Automatic Resource-Aware Layer Placement , Jay H. Park, Sunghwan Kim, Jinwon Lee, Myeongjae Jeon, Sam H. Noh	<i>USENIX NSDI 2019</i>
Paper	Accelerated Training for CNN Distributed Deep Learning through Automatic Resource-Aware Layer Placement , Jay H. Park, Sunghwan Kim, Jinwon Lee, Myeongjae Jeon, Sam H. Noh	<i>arXiv 2019</i>
Poster	Improving Performance of Distributed TensorFlow using CNN Characteristics Exploiting Model Parallelism , Jay H. Park, Jinwon Lee, Sam H. Noh	<i>ACM EuroSys 2018</i>

Extracurricular Activity

Teaching Assistant

Ulsan, S.Korea

TEACHING ASSISTANT, SYSTEM PROGRAMMING

Mar. 2018 - Jun. 2018

- Served as a teaching assistant for the System Programming course, responsible for reviewing and grading student projects, assessing examinations, and providing support for course operations.